

Capacity Guidelines

These guidelines cover capacity planning for the payments platform.

payment-service runs 6 pods in steady state. Each pod holds a database connection pool of 40 connections, for a total of 240. Peak throughput observed is about 1.3 million requests per day.

Scale payment-service to 10 pods when `error_rate_pct` exceeds 2 percent and the connection pool is saturated and there is no recent deploy to roll back. Scaling adds pool capacity and is a temporary mitigation, not a fix.

Do not scale above 12 pods without a database review. Beyond 12 pods the shared database becomes the bottleneck and adding pods increases lock contention.

Redis capacity

The merchant configuration cache uses a single Redis instance sized at 4 GB. Alert at 80 percent memory. The cache TTL is 15 minutes so memory use should be bounded; sustained growth indicates a key that is being written without a TTL and should be investigated.

Never run FLUSHALL against the shared Redis during business hours. Drop individual keys with `cachectl` instead.